

CampusVoice.ai

Closed-Loop Campus Operations Feedback Infrastructure — Hackathon Defense Q&A;

Examiner Briefing Note: This document provides comprehensive, bulletproof answers to the top questions hackathon judges, technical examiners, and campus administrators ask when reviewing CampusVoice. Use this guide during your live presentation, viva Q&A, and technical deep-dive sessions.

SECTION 1: Problem Statement & Core Value Proposition

Q1: What exact problem does CampusVoice solve for large residential campuses like LPU?

Answer: On large residential campuses (e.g., LPU with 35,000+ students and 20+ hostel blocks), a huge **'feedback-to-action gap'** exists. Students raise complaints via informal WhatsApp groups or paper registers, but wardens close tickets unilaterally without fixing the root cause. CampusVoice creates a **Closed-Loop System** where a ticket CANNOT be closed until the student rates the resolution as Satisfied. Unsatisfied ratings trigger automatic 1-click ticket reopening and SLA breach flags.

■ *Key Pitch for Judges: Emphasize 'Closed-Loop Accountability' — Wardens cannot fake resolution because the student holds final verification power.*

Q2: How is CampusVoice different from generic ticketing platforms like Jira, Zendesk, or Google Forms?

Answer: Generic tools lack campus-specific domain logic:

1. **Campus Spatial Context:** Pre-configured with hostel blocks (BH-1, GH-1), room numbers, and dining halls (Mess Alpha).
2. **Weekly Mess Menu Feedback:** Pre-loaded 7-day, 4-meal Indian dish dataset (Paneer Butter Masala, Rajma Chawal) with dish-level rating analytics.
3. **Fake Resolution Visualizer:** Exposes wardens who mark complaints 'Fixed' prematurely.
4. **Public Trust Layer:** Live campus-wide transparency dashboard accessible to all students.

■ *Key Pitch for Judges: Highlight that CampusVoice is purpose-built for campus operational dynamics, not generic corporate helpdesks.*

SECTION 2: System Architecture & Technical Implementation

Q3: Walk us through the technical architecture and tech stack choices.**Answer:**

- **Frontend:** React (Vite) + Tailwind CSS (responsive mobile-first UI with liquid glass styling).
- **Backend:** Node.js + Express REST API providing 20+ modular endpoint routes.
- **Database:** MongoDB with Mongoose ORM for structured models (Users, Complaints, MessMenu, Remarks).
- **Authentication:** JWT (JSON Web Tokens) with Role-Based Access Control (Student, Staff/Warden, Super Admin).
- **Real-time Engine:** Socket.io for instant status change notifications & Chief Warden SLA breach broadcasts.

■ *Technical Tip: Mention that for zero-downtime evaluation, the system includes a synchronous client-side preset layer (`mockData.js`) that runs 100% in-memory if external DB connections time out.*

Q4: How does Role-Based Access Control (RBAC) work in your application?**Answer:** We enforce strict RBAC at both API route and React component levels:

1. **Student:** Lodge complaints, view mess menu, rate meal quality, reopen unsatisfied tickets.
2. **Warden/Staff:** View assigned block complaints, acknowledge tickets, post mandatory action notes, mark resolved.
3. **Super Admin:** Access central gap analytics, SLA breach heatmaps, staff performance leaderboards, export CSV reports.

■ *Technical Tip: Demonstrate the 1-Click Examiner Role Switcher in the top navbar bar to show seamless state transition in 0ms.*

SECTION 3: Key Features & Closed-Loop Workflows**Q5: How does the Automatic Ticket Reopening mechanism operate?****Answer:** When a Warden marks a ticket as 'Resolved', the Student receives an immediate notification to rate the fix (1-5 stars) and select Satisfied/Unsatisfied.

- If **Satisfied:** Ticket moves to final 'Closed' status.
- If **Unsatisfied:** The ticket status automatically flips back to '**Reopened**', increments `reopenedCount`, appends the student's discontent note to the audit trail, and highlights the ticket with an animated purple badge on the Warden dashboard.

■ *Live Demo Step: Show ticket CMP-2026-0005 ('Ceiling fan screeching') which is in 'Reopened' status after student rated 1 star.*

Q6: What are SLA Breaches and how does the system calculate automated escalations?**Answer:** Each ticket is assigned an SLA deadline upon creation based on urgency:

- **Emergency (Water Leak/Power Outage):** 12-Hour SLA Window.
- **High Urgency:** 24-Hour SLA Window.
- **Medium/Low Urgency:** 48 to 72-Hour SLA Window.

If current time exceeds `slaDeadline` before status becomes 'Resolved', the system automatically sets `isSlaBreached = true` and triggers a high-priority red breach banner for the Chief Admin.

■ *Technical Tip: Point out the animated bouncing red SLA Breach badge on tickets where resolution exceeded deadline.*

Q7: How does the Mess Menu & Dining Feedback module work?

Answer: Students can inspect the complete **7-day weekly menu** across all 4 meal slots (Breakfast, Lunch, Evening Snacks, Dinner). Each meal listing displays real dishes (e.g. *Paneer Butter Masala*, *Dal Makhani*, *Chole Bhature*) with live dish rating averages and student reviews. Students can submit dish-specific feedback with photo proof.

■ *Feature Spotlight: Mention that Mess Managers use this data to identify failing dishes and adjust vendor contracts.*

SECTION 4: Scalability, Security & Production Readiness**Q8: How will CampusVoice scale to 50,000+ active students across multiple campus branches?**

Answer:

1. **Database Indexing:** Compound indexes on `{ block: 1, status: 1, createdAt: -1 }` ensure sub-10ms query speeds.
2. **Stateless JWT Auth:** Scales horizontally across serverless functions (Vercel Serverless / AWS Lambda).
3. **CDN Asset Delivery:** Static frontend Vite bundles served via Vercel Edge Network.

■ *Security Note: Input sanitization, password hashing via bcrypt, and strict MIME-type validation on photo uploads.*

Q9: How do you prevent spam or false complaints by students?

Answer:

- **Domain Validation:** Registration requires official university email (`@lpu.in`).
- **Mandatory Room & Block Linkage:** Complaints are pinned to verified student room numbers.
- **Rate Limiting:** Maximum 3 active open complaints per student to prevent flood attacks.

■ *Edge Case Answer: Warden remarks form an immutable append-only audit trail so both sides remain accountable.*

SECTION 5: Future Roadmap & AI/ML Integrations**Q10: What AI/ML models can be integrated into CampusVoice V2?**

Answer:

1. **AI Image Verification:** Computer vision model to verify photo proof before lodging (e.g., detecting electrical spark vs empty room).
2. **NLP Issue Categorization & Priority Scoring:** Natural Language Processing to auto-categorize complaint descriptions and detect urgent keywords ('short circuit', 'water pipe burst').
3. **Predictive Maintenance Analytics:** Time-series forecasting to predict hostel block maintenance cycles before failures occur.

■ *Visionary Pitch: Conclude your presentation by framing CampusVoice as an AI-ready campus infrastructure platform.*